

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 09

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 4 from:</i> • Copying the structural / functional adaptations of living organisms in the design of manufactured items • Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use • Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) • Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) • Termite mound convection copied for natural ventilation of buildings; hollow struted bird bones in lightweight roofs; gecko toe adhesion; burr hooks - Velcro; lotus leaf self-cleaning glass; spider silk for airbags / body armour; bat / dolphin echolocation - ultrasound scanners	4
01.2	• Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores; in controlled doses these affect human physiology • Examples: morphine / codeine from poppies; quinine from cinchona bark (malaria); aspirin from willow bark; taxol from yew (cancers); diosgenin from Mexican yam (contraceptives, cortisone) • Animal examples accepted: AZT from a marine sponge (HIV), shark liver lipids (tumours)	3
01.3	• Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals • Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships)	2
01.4	• Squid giant nerve cells - nerve function / heart disease, stroke; marsupials (kangaroo, wallaby pouch) - embryo development; hippo skin secretion - sunscreen / burns; sea urchin embryos - teratogen testing; armadillo - leprosy; marine sponge proteins - transplant rejection	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	
02.1	• Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion; waste products of one process are raw materials of another so wastes do not build up • Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere • Allowing more plant growth, further stabilising climate and soils	3
02.2	<i>Any 3 from:</i> • Lack of ancient historical / instrumental data • Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions • Lack of sophisticated equipment / inability to measure many factors • Limited coordination between researchers; lack of data collection in many areas (oceans, poles)	3
02.3	• The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell • Air bubbles trapped in ice give the composition of the past atmosphere (CO₂, methane) • Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) • More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	
03.1	• B	1
03.2	<i>Any 4 from:</i> • Deforestation • Ploughing of grassland / expansion of farmland • Reservoir creation / flooding • Mineral extraction, especially open-cast mining • Urban expansion	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	

Question	Marking guidance	Marks
04.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Why needed: growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation; limits to new land and imports • Pest control: cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • Nutrients: natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser; phosphate recycling • Energy: natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • Water: low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water; soil: organic matter maintenance, erosion control • Biodiversity: retaining hedgerows, ditches, ponds, woodlands; CWR conservation and seed banks; agri-environment schemes • Social: control of impacts on rural communities, Fairtrade, local food; consumer choices • Evaluation: yields vs sustainability, cost, need for policy support; judgement	9
04.2	• Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	
05.1	• Yield increases as inputs increase • But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns)	2
05.2	• Second unit: $70 - 50 = 20$ units • Fifth unit: $94 - 90 = 4$ units	2
05.3	• The first units of input on each area give the largest yield increases • Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve	2

Question	Marking guidance	Marks
05.4	- Some areas have much greater potential productivity so investment there is sensible - Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons)	2
05.5	- Yield per unit area is very high because of large inputs - But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	
06.1	- Size: shows whether the population is viable / how urgent action is; sets a baseline to judge success; needed to estimate carrying capacity and sustainable off-take - Distribution: shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors; identifies key areas to protect - Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor - Age structure: many young indicates a growing population; few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous; k-selected species need breeding adults protected	8
06.2	- Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition - Sub-sample counts in known areas scaled up to the whole reserve; mark-release-recapture (Lincoln index) for smaller species	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	
07.1	- Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm²) - They use the polyp's CO₂ and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently - They supply the polyp with glucose and oxygen - over 95% of its food energy; the polyp provides a protected habitat and nutrients - Polyps also trap plankton with cilia and stinging nematocysts; surplus CO₂ is used to build the calcium carbonate skeleton	4
07.2	- High light (shallow, clear water) for the algae; warm stable temperatures (about 18-30 °C); low turbidity; constant salinity; well-oxygenated water - Found only where all these coincide (about 1% of ocean area) - Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance	3
07.3	<i>Any 3 from:</i> - Fisheries: support over a quarter of marine species including fish, lobsters, conch - Erosion protection: complex structure dissipates wave energy protecting low coral islands; coral sand builds shores - Medicinal discoveries: AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts - Climate control: carbonate skeletons sequester CO₂ - Tourism (diving, snorkelling)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.	
08.1	• Rotation: each crop has its own pests; growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early • Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids • Rotation also evens out nutrient demands and includes legumes	4
08.2	• Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6) • It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8 • Cultural control alone loses about 12% yield; integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact	3
08.3	• Large numbers of males are sterilised with gamma radiation and released • Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring; enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly) • Only the target species' females mate with them so non-target species are unaffected; fails if sterilised males behave abnormally or re-colonisation occurs	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.	
09.1	• k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses • Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population • Deep-water aggregations easily located and caught by trawling • Recovery takes decades so fisheries have had to be closed	4
09.2	• Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young • Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed	2
09.3	• Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort • Masks the decline of the breeding population until a sudden collapse	2
09.4	• Neither is a target species but both are caught in demersal trawls; skate now extinct in the Baltic and critically endangered elsewhere • Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	

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10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9
10.2	• A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25